

National University of Computer and Emerging Sciences

Information Retrieval (CS4051)

Date: May 21, 2025 (9AM-12Noon)

Course Instructors

Muhammad Rafi, Basit Jasani

Final Examination

Total Time: 3 Hour

Total Marks: 100

Total Questions: 07

Semester: Spring-2025

Campus: Karachi

Department: AI & DS

Student Name

Roll No

Section

Student Signature

CLO # 1: <Understand the basic concepts and techniques in Information Retrieval.>

Q1. Answer the following questions to the point. [Time: 20 mins] [Marks: 2x5=10]

- What is the future of Information Retrieval in the era of ChatGPT like systems?
- What is an Extended Boolean Model for IR?
- What is an inverted index? What are its components?
- Explain the process of Tokenization. Outline two of its challenges.
- How modern search engine uses a combination of Bi-word index and Position index to efficiently process natural language queries?

CLO # 4: <Appreciate the importance of data structures, such as an index, to allow efficient access to the information in large bodies of text, identify the storage and index requirements for different IR Models>

Q2. Given the following term-document matrix (TF values only): [Time: 20 mins] [Marks: 5+5=10]

Term	D1	D2	D3
data	3	0	1
mining	2	2	0
learning	0	1	3

- Compute the document vectors using TF-IDF weights assuming document frequency for each term is: $df(data)=2$, $df(mining)=2$, $df(learning)=2$, and total documents = 3.
- Compute the cosine similarity between query "data mining" and each document.

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CLO # 2: < 2. Understand how statistical models of text can be used to solve problems in IR, with a focus on how the vector-space model and language models are implemented and applied to document retrieval problems.>

Q3. Answer the following questions.

[Time: 30 mins] [Marks: 4x5=20]

a. Suppose the following scenario:

Original query vector: $Q = [1, 0, 1, 0]$

One relevant document: $D_{rel} = [1, 1, 0, 1]$

One non-relevant document: $D_{nrel} = [0, 0, 1, 1]$

Using Rocchio's formula with $\alpha = 1$, $\beta = 0.75$, $\gamma = 0.15$, compute the modified query vector Q' .

- b. Explain why Rocchio's β (weight for relevant docs) is typically set higher than γ (weight for non-relevant docs). Give an example where this balance matters.
- c. Consider an information need for which there are four relevant documents in the collection. Contrast two systems run on this collection. Their top ten results are judged for relevance as follows (the leftmost item is the top ranked search result):

System1: R N R N N N N R R

System2: N R N N R R R N N

What is the MAP of each system? Which has a higher MAP? Does this result intuitively make sense? What does it say about what is important in getting a good MAP score?

d. What is Normalized Discounted Cumulative Gain? Write down two limitations of it.

CLO # 2: <Understand how statistical models of text can be used to solve problems in IR, with a focus on how the vector-space model and language models are implemented and applied to document retrieval problems>

Q4. Answer the following questions.

[Time: 30 mins] [Marks: 4x5 = 20]

a. What is a Language Model in the context of IR?

b. Consider that you are designing a search system for a medical corpus. Given the following:

- **Query:** $Q = \text{"heart disease risk"}$
- **Document D:** Length ($|D|$) = 120 terms
- **Term Frequencies in D:**
 - $tf(\text{"heart"}, D) = 4$
 - $tf(\text{"disease"}, D) = 0$
 - $tf(\text{"risk"}, D) = 2$
- **Collection Statistics:**
 - Total terms in corpus (C) = 10,000,000

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- Collection Frequency of "heart" ($cf("heart")$) = 8,000
- Collection Frequency of "disease" ($cf("disease")$) = 10,000
- Collection Frequency of "risk" ($cf("risk")$) = 7,500

Compute the smoothed probability of each term in the query ("heart", "disease", "risk") given Document D, using Dirichlet smoothing with $\mu=1500$. The formula for Dirichlet smoothing is:

$$P(t|D) = \frac{tf(t, D) + \mu \cdot P(t|C)}{|D| + \mu}$$

c. What are some of the limitation of Basic Language Model?

d. Consider making a LM from the following training text:

“dil dil pakistan jan jan pakistan hum sab ka pakistan “

- (i) Under a MLE-estimated unigram probability model, what are $P(\text{dil})$ and $P(\text{hum})$?
- (ii) Under a MLE-estimated bigram model, what are $P(\text{pakistan} | \text{dil})$.

CLO # 3: < Understand how statistical models of text can be used for other IR applications, for example clustering, classification and extraction of information >

Q5. Consider the following examples for the task of text classification and answer the following questions: **[Time: 20 mins] [Marks: 5+5=10]**

	docID	words in document	in $c = \text{China?}$
training set	1	Taipei Taiwan	yes
	2	Macao Taiwan Shanghai	yes
	3	Japan Sapporo	no
	4	Sapporo Osaka Taiwan	no
test set	5	Taiwan Taiwan Sapporo	?

- a. Using the **K-Nearest Neighbors (KNN)** with $k=3$, identify the class of test instance **docID=5**.
- b. Using the **Multinomial Naive Bayes** algorithm, estimate the probabilities of each term (feature) used to classify the test instance with **docID = 5**.

CLO # 3: <Understand how statistical models of text can be used for other IR applications, for example clustering, classification and extraction of information.>

Q6. Answer the following questions:

[Time: 20 mins] [Marks: 5+5 = 10]

- a. You are clustering a set of documents based on their content. Some documents are very long, while others are short. Which distance metric would be more appropriate in this scenario: **Euclidean distance** or **cosine similarity**? Justify your choice.

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- b. Given 3 documents represented as tf-idf vectors, perform 2 iterations of k-means clustering with k=2 using D1 and D3 as initial centroids.

D1: [0.4, 0.1, 0.5]

D2: [0.5, 0.0, 0.4]

D3: [0.1, 0.7, 0.2]

CLO # 6: <Understand the issues involved in providing an IR service on a web scale, interactive and neural information retrieval>

Q7. Answer the following questions.

[Time: 30 mins] [Marks: 5+5+10=20]

- a. Outline three differences between **HITS** and **PageRank**.
- b. Define word embedding. How does the Skip-Gram model learn embedding from data?
- c. Consider a graph with 4 nodes: A, B, C, and D. The connections are as follows:
 - A is directly connected to B and D.
 - C is connected to B and D.
 - B is connected to D.

Based on the information provided, answer the following:

- (i) Draw the given graph.
- (ii) Compute the link matrix (M) for the given graph.
- (iii) With a damping factor $d=0.65$ compute the Google PageRank matrix G, here E is an Identity Matrix of the same order. N is the number of pages 4 in the given case. Compute one iteration of the equation $PR^{(1)} = G * PR^{(0)}$ starting from $PR^{(0)} = [\frac{1}{4} \frac{1}{4} \frac{1}{4} \frac{1}{4}]^T$

$$G = dM + (1 - d) \cdot \frac{1}{N} \cdot E$$